

Power analysis: optimisation of sea lice monitoring programme

Abundance

Bela Klimesova

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1 Preparation of sea lice dataset

R packages necessary of data analysis

```
library(ggplot2)
library(glmmTMB)
library(progress)
suppressMessages(library(dplyr))
suppressMessages(library(future.apply))
library(progressr)
```

1.1 Sea lice dataset

Sea lice monitoring dataset provided by the industry producer

```
Industry <- read.table(  
  "Industry.csv",  
  header = TRUE,  
  sep = ',',  
  fill = TRUE,  
  dec = ".",  
  na.strings = "NA"  
)
```

2 Empirical data analysis

2.1 Modelling of temporal increase of sea lice abundance levels

Adjustment of the variable pen: unique identification for each farm

```
Industry$Pen_un <- paste(Industry$SiteName, Industry$Pen.Count, sep = "_")
```

GLMM model abundance increase per production week, random effects to take into account the hierarchical structure of the data

```
mod_A <- glmmTMB(  
  Adult.female ~ Week_Production + (1 | SiteName) + (1 | Pen_un) + (1 | Year_Production),  
  family = nbinom2,  
  data = Industry  
)
```

Exploration of the model

```
# Results overview  
summary(mod_A)
```

```
## Family: nbinom2 ( log )  
## Formula:  
## Adult.female ~ Week_Production + (1 | SiteName) + (1 | Pen_un) +  
## (1 | Year_Production)  
## Data: Industry  
##  
##      AIC      BIC    logLik -2*log(L)  df.resid  
## 168003.3 168062.2 -83995.6  167991.3    136121  
##  
## Random effects:  
##  
## Conditional model:  
## Groups      Name      Variance Std.Dev.  
## SiteName    (Intercept) 2.42490  1.5572  
## Pen_un      (Intercept) 0.06956  0.2637
```

```
## Year_Production (Intercept) 2.07113 1.4391
## Number of obs: 136127, groups: SiteName, 9; Pen_un, 96; Year_Production, 7
##
## Dispersion parameter for nbinom2 family (): 0.397
##
## Conditional model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -4.900682  0.770641  -6.36 2.03e-10 ***
## Week_Production 0.071857  0.000605 118.78 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# AIC
AIC(mod_A)
```

```
## [1] 168003.3
```

```
# Collinearity
performance::model_performance(mod_A)
```

```
## # Indices of model performance
##
## AIC      |      AICc |      BIC | R2 (cond.) | R2 (marg.) |  ICC |  RMSE
## -----|-----|-----|-----|-----|-----|-----
## 1.680e+05 | 1.680e+05 | 1.681e+05 | 0.827 | 0.267 | 0.764 | 3.550
##
## AIC      | Sigma | Score_log | Score_spherical
## -----|-----|-----|-----
## 1.680e+05 | 0.397 | -0.718 | 0.002
```

2.2 Exploration of temporal increases in infection levels

2.2.1 Weekly increase for all farms

Rates ratio increase of abundance converted to %

```
slope <- fixef(mod_A)$cond["Week_Production"]
RR <- exp(slope)
per_increase <- (RR - 1) * 100
per_increase
```

```
## Week_Production
## 7.450124
```

2.2.2 Weekly increase for each farm separately

GLMM of temporal increase for each farm

```
mod_A1 <- glmmTMB(
  Adult.female ~ Week_Production + (Week_Production |
                                     SiteName) +
  (1 | Pen_un) + (1 | Year_Production),
  family = nbinom2,
  data = Industry
)
```

Calculation of percentage increase, rates ratio increase of abundance converted to %

```
beta_fixed <- fixef(mod_A1)$cond["Week_Production"]
ran <- ranef(mod_A1)$cond$SiteName
site_dev <- ran[, "Week_Production"]

site_slopes <- beta_fixed + site_dev
RR_sites <- exp(site_slopes)
perc_inc_sites <- (RR_sites - 1) * 100

results <- data.frame(
  Site = rownames(ran),
  Slope = site_slopes,
  Rate_Ratio = RR_sites,
  Percent_Increase = perc_inc_sites
)

results
```

##	Site	Slope	Rate_Ratio	Percent_Increase
## 1	a	0.04220941	1.043113	4.311289
## 2	b	0.08563945	1.089413	8.941347
## 3	c	0.05987700	1.061706	6.170595
## 4	d	0.06912583	1.071571	7.157104
## 5	e	0.04920923	1.050440	5.044011
## 6	f	0.10818473	1.114254	11.425356
## 7	g	0.08456168	1.088240	8.823997
## 8	h	0.05641252	1.058034	5.803405
## 9	i	0.03191718	1.032432	3.243199

2.2.3 Abundance variation between two consecutive inspections

Comparing abundance levels between two consecutive visits in raw data

Abundance at farm level

```
Ind1 <- aggregate(
  Industry$Adult.female,
  list(
    Industry$SiteName,
    Industry$Year_Production,
    Industry$Week_Production,
    Industry$Count.Date
  ),
```

```

    mean
  )
colnames(Ind1) <- c('Site', 'Year', 'Week', 'Date', 'Abun')

Ind1$Date <- as.Date(Ind1$Date, format = "%Y-%m-%d")
Ind1_s <- Ind1[order(Ind1$Date), ]

```

Calculation of percentage change in abundance levels

```

Ind1_s <- Ind1_s %>%
  arrange(Site, Date) %>%
  group_by(Site) %>%
  mutate(Abun_lag = lag(Abun),
         Increase = ifelse(Abun_lag == 0 | is.na(Abun_lag), NA,
                           ((Abun - Abun_lag) / Abun_lag) * 100)) %>%
  ungroup()

Ind1_s$Increase <- round(Ind1_s$Increase)

```

Exploration of abundance change

```

Ind1_s1 <- as.data.frame(Ind1_s)

# Increase
summary(Ind1_s1[Ind1_s1$Increase>0,]$Increase)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##      1.00   35.25   93.00  203.79  209.25 4850.00    479

```

```

# Decrease
summary(Ind1_s1[Ind1_s1$Increase<0,]$Increase)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
## -100.00 -100.00  -58.00  -59.75  -31.00   -1.00    479

```

Increase in abundance from 0

```

Ind_0<-Ind1_s[Ind1_s$Abun_lag == 0, ]
Ind_0 <- as.data.frame(Ind_0)
Ind_0<-Ind_0[!(is.na(Ind_0$Year)),]

nrow(Ind_0[Ind_0$Abun>0.19,])/nrow(Ind_0)

```

```
## [1] 0.008510638
```

2.3 Estimated parameters from GLMM

Slope: rate of abundance increase

```
slope <- fixef(mod_A)$cond["Week_Production"]
```

Mean0: intercept on the log scale

```
intercept <- fixef(mod_A)$cond["(Intercept)"]  
mean0 <- exp(intercept)
```

Theta

```
theta <- 1 / (sigma(mod_A)^2)
```

Random effects sd

```
vc <- VarCorr(mod_A)$cond  
  
site_sd <- as.numeric(attr(vc$SiteName, "stddev"))  
pen_sd <- as.numeric(attr(vc$Pen_un, "stddev"))  
year_sd <- as.numeric(attr(vc$Year_Production, "stddev"))
```

Number of fish per pen

```
fish_per_pen <- Industry %>%  
  group_by(SiteName, Date, Pen.ID) %>%  
  summarise(n_fish = n_distinct(FishID), .groups = "drop") %>%  
  summarise(mean_fish = mean(n_fish)) %>%  
  pull(mean_fish) %>%  
  round()
```

Number of sites and years

```
n_sites <- length(unique(Industry$SiteName))  
n_years <- length(unique(Industry$Year_Production))
```

3 Simulation-based power analysis on national level

3.1 Power analysis function

Function for simulation-based power analysis

```
simulate_test_s <- function(site_sd,  
                             pen_sd,  
                             year_sd,  
                             n_sites,  
                             n_pens,  
                             n_years,  
                             fish_per_pen,  
                             n_visits,  
                             mean0,  
                             slope,
```

```

        theta,
        n_reps = 100) {
intercept <- log(mean0) # Convert baseline abundance to log scale

# Run simulation
power_results <- replicate(n_reps, { # run the simulation n_reps times

  # random effect
  site_re <- rnorm(n_sites, 0, site_sd)
  pen_re <- rnorm(n_sites * n_pens, 0, pen_sd)
  year_re <- rnorm(n_years, 0, year_sd)

  # study design
  dat <- expand.grid(
    Site = factor(1:n_sites),
    Pen = factor(1:n_pens),
    Week = 1:n_visits,
    Year = factor(1:n_years)
  )

  # random effect
  dat$Pen_un <- interaction(dat$Site, dat$Pen)
  pen_levels <- interaction(dat$Site, dat$Pen)

  dat$site_re <- site_re[dat$Site]
  dat$pen_re <- pen_re[as.numeric(pen_levels)]
  dat$year_re <- year_re[dat$Year]

  # Expand to fish-level data
  dat_fish <- dat[rep(seq_len(nrow(dat)), each = fish_per_pen), ]
  dat_fish$FishID <- rep(1:fish_per_pen, times = nrow(dat))

  # Linear predictor
  eta <- intercept + slope * dat_fish$Week +
    dat_fish$site_re + dat_fish$pen_re + dat_fish$year_re
  mu <- exp(eta)

  # Generate counts per fish
  dat_fish$Adult.female <- rnbino(nrow(dat_fish), mu = mu, size = theta)

  # Fit model
  formula <- "Adult.female ~ Week"

  # Add random effects only if more than 1 level exists
  if (n_sites > 1)
    formula <- paste0(formula, " + (1|Site)")
  if (n_pens > 1)
    formula <- paste0(formula, " + (1|Pen_un)")
  if (n_years > 1)
    formula <- paste0(formula, " + (1|Year)")

  m <- try(glmmTMB(as.formula(formula), family = nbinom2, data = dat_fish),
    silent = TRUE)

```

```

if (inherits(m, "try-error"))
  return(NA) # If model fails to converge, skip this replicate
if (!m$sdr$pdHess)
  return(NA) # skip non-converged models

# Detection rule
est <- summary(m)$coefficients$cond["Week", ]
est["Estimate"] > 0 && est["Pr(>|z|)"] < 0.05
})

# Remove NAs (failed fits)
power_clean <- power_results[!is.na(power_results)]

# Calculate power and confidence interval
power_estimate <- mean(power_clean)
ci <- binom.test(sum(power_clean), length(power_clean))$conf.int

list(
  power = power_estimate,
  lower_CI = ci[1],
  upper_CI = ci[2],
  n_reps = length(power_clean)
)
}

```

3.2 Sampling scenarios

Rates ratios (effect size)

```
rate_ratios <- c(1.04, 1.06, 1.1, 1.2, 1.3)
```

Number of fish per pen

```
fish_vec <- c(5, 10, 15, 20, 25, 30)
```

Number of visits

```
visit_vec <- c(2, 3, 4, 6, 8)
```

Number of pens

```
pen_vec <- c(2, 5, 8, 10, 12, 15)
```

Replicates

```
n_reps <- 1000
```


3.3 Simulation run

In the following text the code to run simulation is provided, however, results are uploaded separately.

Create design grid for all scenarios

```
design_grid <- expand.grid(  
  RR = rate_ratios,  
  fish_per_pen = fish_vec,  
  n_visits = visit_vec,  
  n_pens = pen_vec  
)
```

Prepare columns for results

```
design_grid$power <- NA_real_  
design_grid$lower_CI <- NA_real_  
design_grid$upper_CI <- NA_real_  
design_grid$n_success <- NA_integer_
```

Set parallel backend

```
# plan(multisession)
```

Reproducibility

```
# set.seed(123)
```

Progress bar

```
# handlers(global = TRUE)  
# handlers("txtprogressbar")
```

Simulation

```
# with_progress({  
#  
#   p <- progressr::progressor(steps = nrow(design_grid))  
  
#   results_list <- future_lapply(seq_len(nrow(design_grid)), function(i) {  
  
#     Convert RR to slope  
#     slope_i <- log(design_grid$RR[i])  
  
#     Run simulation  
#     sim_res <- simulate_test_s(  
#       site_sd = site_sd,  
#       pen_sd = pen_sd,  
#       year_sd = year_sd,  
#       n_sites = n_sites,  
#       n_pens = design_grid$n_pens[i],  
#       n_years = n_years,
```

```

#   fish_per_pen = design_grid$fish_per_pen[i],
#   n_visits = design_grid$n_visits[i],
#   mean0 = mean0,
#   slope = slope_i,
#   theta = theta,
#   n_reps = n_reps
# )

#   p()

#   # Return results
#   list(
#     power = sim_res$power,
#     lower_CI = sim_res$lower_CI,
#     upper_CI = sim_res$upper_CI,
#     n_success = sim_res$n_reps
#   )

# }, future.seed = TRUE)

# })

```

Adding results to prepared dataset

```

# design_grid$power      <- sapply(results_list, `[[`, "power")
# design_grid$lower_CI   <- sapply(results_list, `[[`, "lower_CI")
# design_grid$upper_CI   <- sapply(results_list, `[[`, "upper_CI")
# design_grid$n_success  <- sapply(results_list, `[[`, "n_success")

```

Calculate increase [%] from RR

```

# design_grid$perc_increase <- round((design_grid$RR - 1) * 100)

```

3.3.1 Results of simulation

Loading saved results of simulation

```

design_grid <- readRDS("Abundance_National.rds")

```

3.4 Results visualisation

Substitute 1 for 0.999

```

design_grid1 <- design_grid
if (any(design_grid1$power == 1, na.rm = TRUE)) {
  design_grid1$power[design_grid1$power == 1] <- 0.999
}

```

Final plot for abundance on national level

```

ggplot(design_grid1, aes(x = fish_per_pen, y = n_pens, z = power)) +
  geom_contour_filled(breaks = seq(0, 1, by = 0.1), alpha = 1) +
  geom_contour(breaks = 0.8,
              color = "black",
              linewidth = 1.1) +
  facet_grid(
    perc_increase ~ n_visits,
    labeller = labeller(
      n_visits = function(x)
        paste(x, "visits"),
      perc_increase = function(x)
        paste0(x, "% increase")
    )
  ) +
  scale_fill_viridis_d(name = "Power", option = "A", begin = 0.15, end = 0.9) +
  labs(
    x = "Number of fish sampled per pen",
    y = "Number of pens"
  ) +
  theme_minimal(base_size = 10) +
  theme(panel.grid = element_blank(), strip.text = element_text(face = "bold"))

```

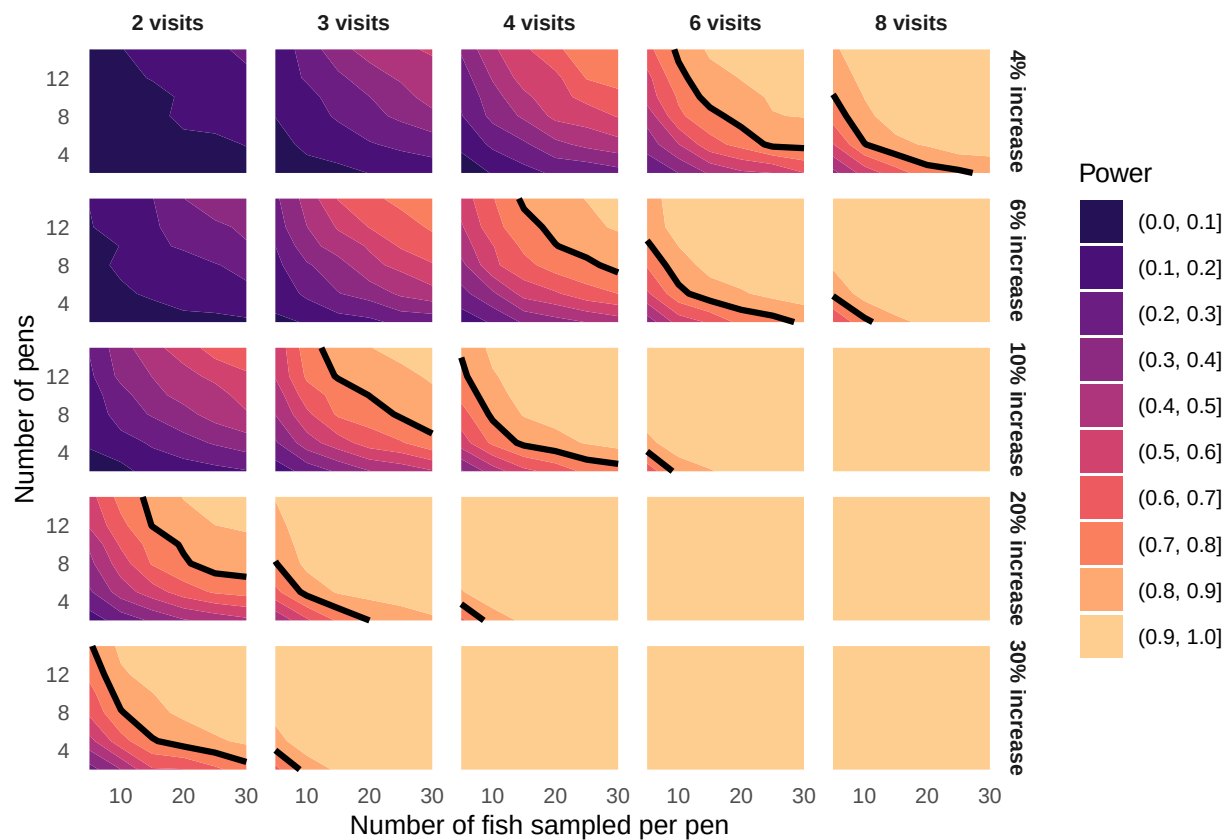


Figure 1: Power analysis for abundance on national level

Minimal sample size for each sampling scenario

Table 1: Minimal sample size for abundance at national level

perc_increase	n_visits	n_pens	fish_per_pen	No_Fish	power
4	6	5	25	750	0.82 [0.8, 0.85]
4	8	2	30	480	0.82 [0.8, 0.85]
6	4	15	15	900	0.82 [0.8, 0.84]
6	6	2	30	360	0.82 [0.79, 0.84]
6	8	5	5	200	0.82 [0.79, 0.84]
10	3	12	15	540	0.81 [0.78, 0.83]
10	4	5	15	300	0.82 [0.8, 0.85]
10	6	2	10	120	0.84 [0.81, 0.86]
10	8	2	5	80	0.9 [0.88, 0.92]
20	2	12	15	360	0.8 [0.78, 0.83]
20	3	2	20	120	0.8 [0.77, 0.83]
20	4	2	10	80	0.85 [0.82, 0.87]
20	6	2	5	60	0.96 [0.95, 0.97]
20	8	2	5	80	1 [0.99, 1]
30	2	5	20	200	0.85 [0.82, 0.87]
30	3	2	10	60	0.84 [0.81, 0.86]
30	4	2	5	40	0.9 [0.88, 0.92]
30	6	2	5	60	1 [0.99, 1]
30	8	2	5	80	1 [1, 1]

```

best_scenarios <- design_grid %>%
  mutate(No_Fish = fish_per_pen * n_visits * n_pens) %>%
  filter(power >= 0.8) %>%
  group_by(perc_increase, n_visits) %>%
  arrange(No_Fish, n_visits, n_pens, .by_group = TRUE) %>%
  slice(1) %>%
  ungroup() %>%
  arrange(perc_increase, n_visits, n_pens, No_Fish) %>%
  as.data.frame()

Best<-best_scenarios[,c(9,3,4,2,10,5)]
Best[,6]<-paste0(round(best_scenarios$power,2), ' ', round(best_scenarios$lower_CI,2), ' ',
  round(best_scenarios$upper_CI,2), ' ')

```

```
knitr::kable(Best)
```

4 Simulation based-power analysis on farm level

4.1 Sampling scenarios

Farm level: 1 farm per 1 production year

```

n_sites <- 1
n_years <- 1

```

Rates ratio (effect size)

```
rate_ratios <- c(1.2, 1.3, 1.5, 2, 2.5)
```

Number of fish

```
fish_vec <- c(5, 10, 15, 20, 25, 30)
```

Number of visits

```
visit_vec <- c(4, 6, 8, 10, 12)
```

Number of pens

```
pen_vec <- c(2, 5, 8, 10, 12, 15)
```

Replicates

```
n_reps <- 1000
```

4.2 Simulation run

In the following text the code to run simulation is provided, however, results are uploaded separately.

Create design grid for all scenarios

```
design_grid <- expand.grid(  
  RR = rate_ratios,  
  fish_per_pen = fish_vec,  
  n_visits = visit_vec,  
  n_pens = pen_vec  
)
```

Prepare columns for results

```
design_grid$power <- NA_real_  
design_grid$lower_CI <- NA_real_  
design_grid$upper_CI <- NA_real_  
design_grid$n_success <- NA_integer_
```

Set parallel backend

```
# plan(multisession)
```

Reproducibility

```
# set.seed(123)
```

Progress bar

```
# handlers(global = TRUE)
# handlers("txtprogressbar")
```

Simulation

```
#with_progress({
#
#  p <- progressr::progressor(steps = nrow(design_grid))

#  results_list <- future_lapply(seq_len(nrow(design_grid)), function(i) {

    # Convert RR to slope
    #  slope_i <- log(design_grid$RR[i])

    # Run simulation
    #  sim_res <- simulate_test_s(
    #    site_sd = site_sd,
    #    pen_sd = pen_sd,
    #    year_sd = year_sd,
    #    n_sites = n_sites,
    #    n_pens = design_grid$n_pens[i],
    #    n_years = n_years,
    #    fish_per_pen = design_grid$fish_per_pen[i],
    #    n_visits = design_grid$n_visits[i],
    #    mean0 = mean0,
    #    slope = slope_i,
    #    theta = theta,
    #    n_reps = n_reps
    #  )

    #    p()

    # Return results
    #  list(
    #    power = sim_res$power,
    #    lower_CI = sim_res$lower_CI,
    #    upper_CI = sim_res$upper_CI,
    #    n_success = sim_res$n_reps
    #  )

    #  }, future.seed = TRUE)

#})
```

Adding results to a prepared dataset

```
# design_grid$power      <- sapply(results_list, `[[`, "power")
# design_grid$lower_CI   <- sapply(results_list, `[[`, "lower_CI")
# design_grid$upper_CI   <- sapply(results_list, `[[`, "upper_CI")
# design_grid$n_success  <- sapply(results_list, `[[`, "n_success")
```

Calculate percent increase from RR for clarity

```
# design_grid$perc_increase <- (design_grid$RR - 1) * 100
```

4.2.1 Results of simulation

Loading saved results of simulation

```
design_grid <- readRDS("Abundance_Farm.rds")
```

4.3 Results visualisation

Substitute 1 for 0.999 due to visualization

```
design_grid1 <- design_grid
if (any(design_grid1$power == 1, na.rm = TRUE)) {
  design_grid1$power[design_grid1$power == 1] <- 0.999
}
```

Final plot for abundance on national level

```
ggplot(design_grid1, aes(x = fish_per_pen, y = n_pens, z = power)) +
  geom_contour_filled(breaks = seq(0, 1, by = 0.1), alpha = 1) +
  geom_contour(breaks = 0.8,
              color = "black",
              linewidth = 1.1) +
  facet_grid(
    perc_increase ~ n_visits,
    labeller = labeller(
      n_visits = function(x)
        paste(x, "visits"),
      perc_increase = function(x)
        paste0(x, "% increase")
    )
  ) +
  scale_fill_viridis_d(name = "Power", option = "A", begin = 0.15, end = 0.9) +
  labs(
    x = "Number of fish sampled per pen",
    y = "Number of pens"
  ) +
  theme_minimal(base_size = 10) +
  theme(panel.grid = element_blank(), strip.text = element_text(face = "bold"))
```

Calculation of minimal sample size

```
best_scenarios <- design_grid %>%
  mutate(No_Fish = fish_per_pen * n_visits * n_pens) %>%
  filter(power >= 0.8) %>%
  group_by(perc_increase, n_visits) %>%
  arrange(No_Fish, n_visits, n_pens, .by_group = TRUE) %>%
  slice(1) %>%
  ungroup() %>%
```

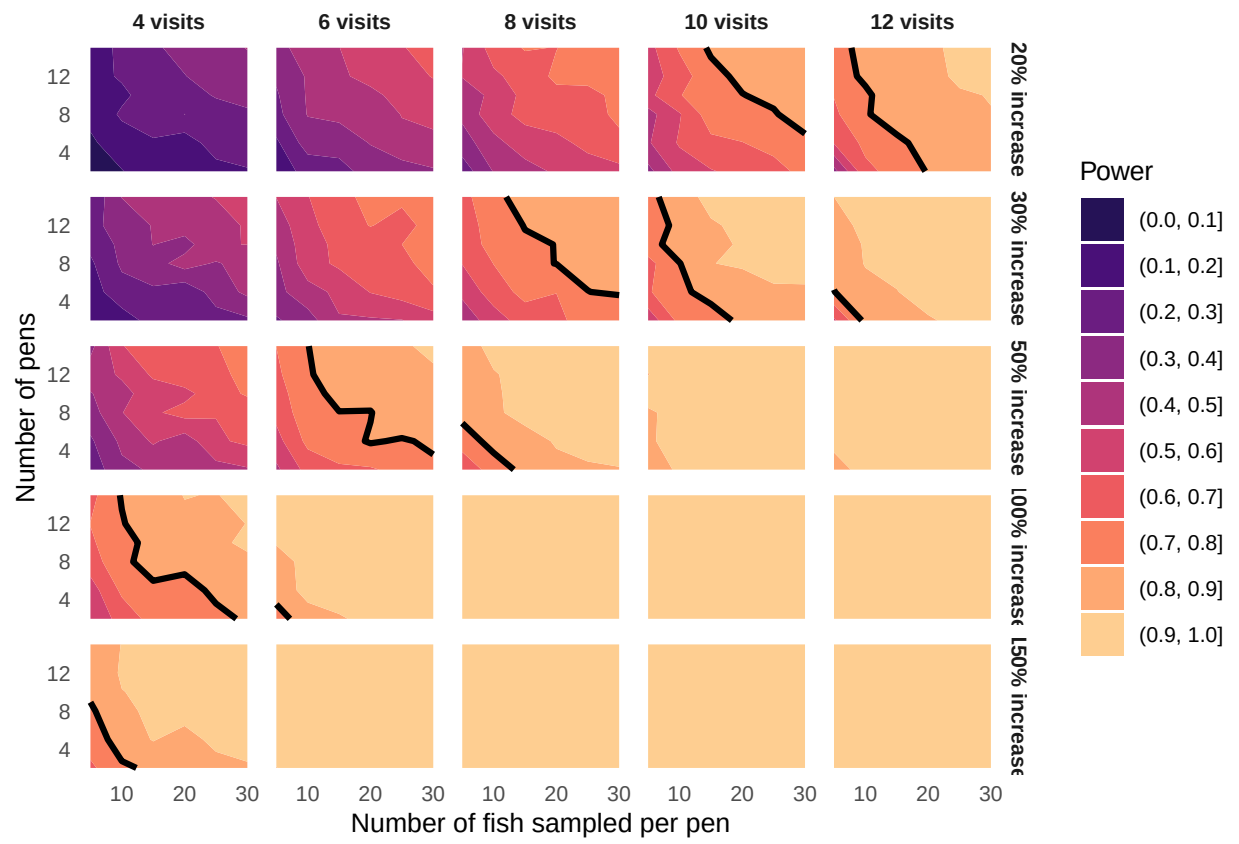


Figure 2: Power analysis for abundance at farm level

Table 2: Minimal sample size for abundance at farm level

perc_increase	n_visits	n_pens	fish_per_pen	No_Fish	power
20	10	15	15	2250	0.82 [0.79, 0.84]
20	12	2	20	480	0.81 [0.77, 0.84]
30	8	5	30	1200	0.81 [0.77, 0.84]
30	10	2	20	400	0.82 [0.78, 0.85]
30	12	2	10	240	0.83 [0.8, 0.86]
50	6	5	20	600	0.81 [0.78, 0.84]
50	8	2	15	240	0.82 [0.79, 0.85]
50	10	2	5	100	0.81 [0.77, 0.84]
50	12	2	5	120	0.84 [0.81, 0.87]
100	4	2	30	240	0.83 [0.8, 0.86]
100	6	2	10	120	0.86 [0.83, 0.89]
100	8	2	5	80	0.92 [0.9, 0.94]
100	10	2	5	100	0.98 [0.97, 0.99]
100	12	2	5	120	1 [0.99, 1]
150	4	2	15	120	0.81 [0.78, 0.85]
150	6	2	5	60	0.9 [0.87, 0.92]
150	8	2	5	80	0.98 [0.97, 0.99]
150	10	2	5	100	1 [0.99, 1]
150	12	2	5	120	1 [1, 1]

```

arrange(perc_increase, n_visits, n_pens, No_Fish) %>%
as.data.frame()

```

```

Best<-best_scenarios[,c(9,3,4,2,10,5)]
Best[,6]<-paste0(round(best_scenarios$power,2), ' ',round(best_scenarios$lower_CI,2), ' ', ' ',
                round(best_scenarios$upper_CI,2), ' '))

```

```
knitr::kable(Best)
```